Model environments for integration and support

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Reference models accurately reflect live networks — increasingly this means they are more complex, have shorter development cycles, and are expected to deliver more value-for-money than ever before. Fortunately, it is precisely these real-world issues that reference models are designed to address. This paper examines the evolving role of reference models, and demonstrates that they are an increasingly vital element in BTs integration strategy, facilitating the delivery of new services within time and budget. A few of the models within the integration facility are examined to give some idea of their scope.

1. Introduction

Reference models have been at the core of BT's network integration strategy for several decades. A world-leading facility has been built at BT Laboratories (BTL), yet its existence is almost unknown outside the immediate testing community. Indeed, most people have little idea what a reference model is.

To clarify this point first, a reference model is a collection of hardware and/or software, configured and managed to closely mirror the operation of the corresponding live network or service. Of course, almost every new service may already have a model of some sort, normally used for development purposes — but these are not reference models.

Development systems are not usually maintained under strict configuration management control — by definition, in a development area, all sorts of new ideas are being evaluated. Once a service has been rolled out, then the development team will look ahead to the next release, and their model may cease to bear any relationship to the existing operational service.

True reference models are an accurate reflection of the service in operation at that time. (This is not necessarily true immediately before a network upgrade, where the new release will be undergoing pre-launch testing on the model.)

BT recognises that rapid delivery of right-first-time services requires a heavy investment in testing, and that testing is only really possible when adequate reference model facilities exist. There are significant savings to be made if these facilities are co-located.

The integration facility is at the heart of BT's integration strategy. Here, at a single site, lies the greatest concentration of telephony reference models in the UK, and quite possibly the world.

Traditionally, core public switched telephone network (PSTN) activities have dominated BT's business, and integration testing has therefore been biased in this direction. Adequate testing of PSTN networks requires a significant set of models, and the majority of these are housed in the integration facility (IF) at BTL. Over the last decade however, while core PSTN models have remained a significant component of the IF, new models are forming an increasing percentage of the expanding total.

So what are reference models used for? Section 2 details some of the potential uses, but each model is unique, and the precise functions they perform will vary both between models, and with time. The main use of the models in the IF is (not surprisingly) integration testing, and the requirements for this activity are paramount when the models are being designed and upgraded.

2. Integration facility

The name 'integration facility' refers not only to the building which houses many of the true reference models within BT (and most of those related to the PSTN core network), but also refers more generically to the building plus the equipment and the people who manage it. Both meanings are used interchangeably within this document.

Some models are located remotely from the IF, mostly for historic reasons, but also due to the fact that new models are arriving much faster than older ones are disappearing. A further reason for having models remote to the IF is the difficulty of managing local versions of major operational platforms such as the customer service system (CSS). For systems such as this, partitions are created on test environments managed as part of the live network, and these partitions form part of a logical model, based on the IF.

2.1 Background

The integration facility building is situated at BTL and it was originally built in the late 1970s to house System X development equipment, then being done by Switching Systems division. It was designed from the outset to house telephone exchanges, although since then the facility has incorporated a number of new technologies, including Switched Multimegabit Data Service (SMDS), asynchronous transfer mode (ATM), and Internet protocol (IP). Almost 5000 square metres of floor space are currently dedicated to models.

In deference to the previous use of the BTL site as an aircraft-testing establishment, all the main models are named after aircraft that were tested at Martlesham Heath between the years of 1917 and 1962. Familiar names include Harrier, Mustang, and Spitfire.

The IF comprises a number of different models, which have been grouped into the categories listed below. These categories have been chosen to align, where practical, with standard BT platform definitions.

In addition to the models themselves, the IF must provide facilities for testers to access the models. This is done using test bays — each test bay is equipped with a number of interfaces, allowing the testers to gain access to whatever models they need, without actually having to enter (secure) equipment areas. This is a vital aspect of ensuring that models remain under strict configuration management control.

Test bays (see Fig 1) vary in composition, with different interfaces being provided depending on the primary purpose of that test bay. For example, some are dedicated to UK core testing, in which case they have a large number of PSTN interfaces, whereas the multimedia test bays are mainly equipped with local area network (LAN) connections.

2.2 Models

This section describes some of the models within the IF. This is not an exhaustive list, but serves to demonstrate the breadth of coverage available. The diagrams that have been included are meant to illustrate the basic structure of a model, or perhaps to give particular emphasis to one aspect of that model. They are not intended to describe a model in detail — which would require rather more space than is available.

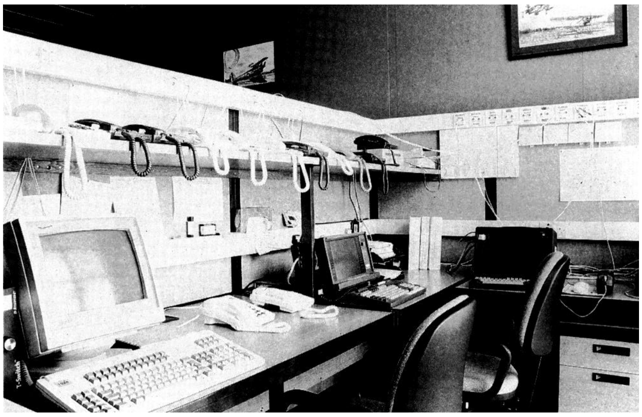

Fig 1 Typical test bay.

UK core switching

This category has traditionally formed the mainstay of the IF. UK core switching is essentially represented by a number of telephone exchanges [1], together with their appropriate management systems. At least one instance of every type of telephone exchange currently in the BT network is present in the IF. A selection of the exchanges present includes those listed in Table 1.

Table 1 Sample exchanges in the integration facility.

Exchange	Name	Function
System X DMSU(GPT) (see Fig 2)	Vulcan	UK PSTN & ISDN
System X DMSU Mobile (GPT)	Gipsy	UK PSTN & ISDN
System X Local (GPT)	Harrier, Bulldog	UK Local PSTN & ISDN
AXE10 local (ETL)	Blenheim, Buzzard	UK Local PSTN & ISDN
AXE10 local mobile (ETL)	Falcon	UK Local PSTN & ISDN
5ESS ISC (AT&T)	Mustang	International Network & DSN
UXD5 (BT)	Shackleton, Spitfire	Rural Exchange
TXE4 (Nortel)	1A, 1B, 1C	UK Local PSTN

Most of the exchanges are interlinked, so that the IF provides the same potential connectivity that exists in the core BT network. Thus it is possible to test (almost) any configuration that could possibly be encountered in the field. This is made possible by a series of frame rooms (giant patch panels), and a few thousand kilometres of cable. This infrastructure is very much taken for granted by those who use the IF, but without it, rapid reconfigurations would be impossible. This is one reason why it is preferable to co-locate the exchanges. Interconnect between the

Fig 2 Vulcan System X DMSU model.

different exchanges is becoming even more problematic, with the growth of newer technologies like ATM. Due to the high data rates involved, and the corresponding stringent distance limitations, care must be taken even when siting equipment within the same building — otherwise costly distance extenders are required.

All these exchanges are managed by a model operations and maintenance centre (OMC) system (TFOMC2), mirroring the situation that would exist in the field.

The selection of exchanges is constantly under review to ensure it continues to reflect the actual network — Strowger equipment no longer figures in the list, and TXE4 will be phased out once the core network has been upgraded fully. Currently, the model TXE4 resides at New Southgate, London, and is maintained by Nortel on behalf of the IF [2]. The TXE4 is fully integrated with the rest of the IF, and numerous E1 links are provided back to the IF exchanges.

One example of the use of the core switching platforms is that of number portability [3, 4]. This is a major programme to meet regulatory demands to allow customers easy means to transfer their number between telephone companies. The IF played a central role in establishing the test environment for this programme, but relatively little of the equipment actually resides within the IF. The network diagram is shown in Fig 3.

All of the elements in Fig 3 shown above the ‘captive PSTN’ elements are remote to the IF — indeed they are distributed throughout the UK. However, when taken logically together, they form a single integrated facility, based around the IF. This kind of role may be more common in future, as the use of large, centralised systems (especially for customer management and billing) continues to dominate.

UK core transmission

Underlying most of the switching models are the transmission models. Originally this was the plesiochronous digital hierarchy (PDH), but this is increasingly being replaced by the synchronous digital hierarchy (SDH) (see Fig 4) and ATM models. The PDH model allows testing on the entire ‘mux mountain’, all the way up from 2 Mbit/s to 565 Mbit/s, including the various co-axial and fibre line systems involved.

PDH is being replaced by two SDH network layers. This comprises the international Ericsson/Marconi SDH network (known as Focus GN) which models the access points for the international traffic coming into the UK via optical and electrical cables (TAT 12/13, CANTAT, etc) and the satellite stations at Goonhilly and Madley. The GPT SDH inland network then transports traffic within the UK. The models of both the SDH networks include crossconnects and add/drop multiplexers, again with all the attendant line systems. The model also includes the relevant management systems, replicating both a network administrative computer centre (NACC) and a network operations unit (NOU).

There are two ATM models — one for BT services and one for Concert services. The Concert ATM model is covered later in this section under ‘Global transmission’. The BT ATM model [5] consists of two Newbridge core ATM crossconnects, and associated equipment. ATM is used in two principal configurations in the BT network, and it was not cost-effective to model each independently. Hence, the ATM model can be reconfigured to support either application. The first is a two-node inland ATM network (Fig 5) interconnected via the GPT SDH model equipment. The second configuration (Fig 6) is an international ATM network, using the Ericsson SDH model.

The KiloStream model (see Fig 7) is comprised of five crossconnects, eleven intelligent muxes (ENA1As), four multiplexers and various types of network terminating units. These are controlled by the order processing platform (GENETIC) which downloads the appropriate commands to the GPT and AT&T element management systems (RENACE and I2000 respectively). Other models, e.g. Customer Oriented System for Management of Special Services (COSMOSS), based in Ipswich, interface to the KiloStream model to enable the entire software delivery process to be modelled.

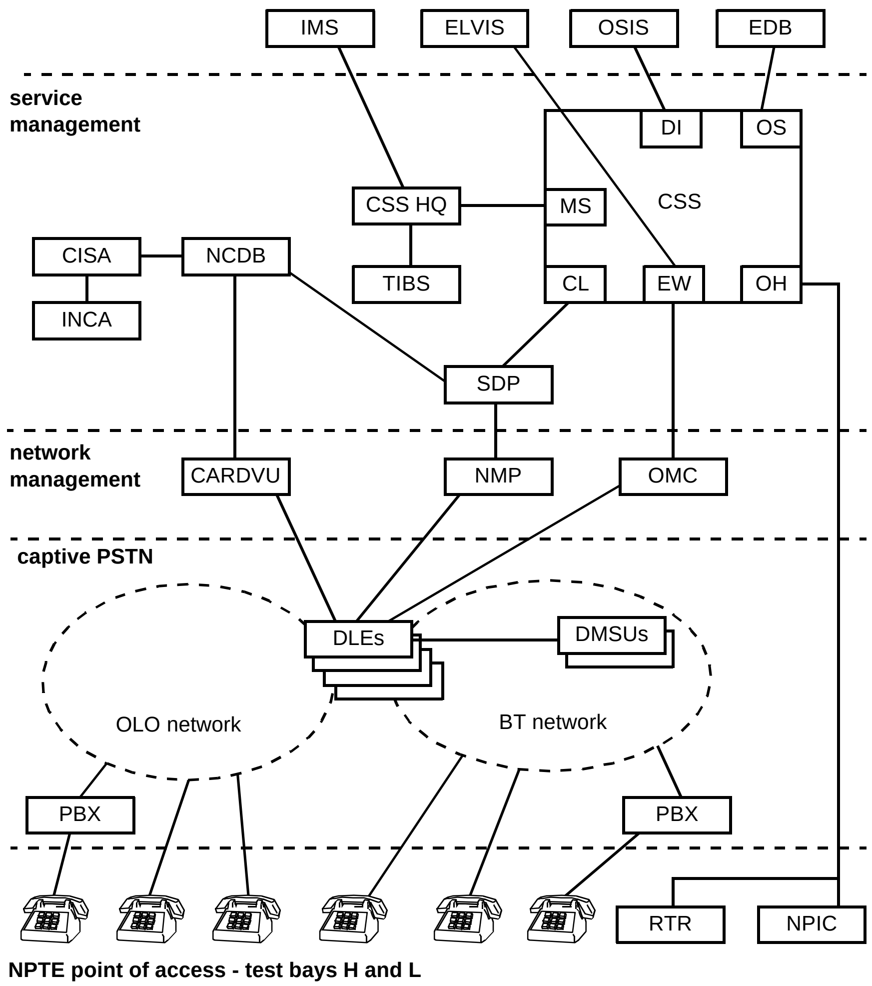

Fig 3 Number portability test environment.

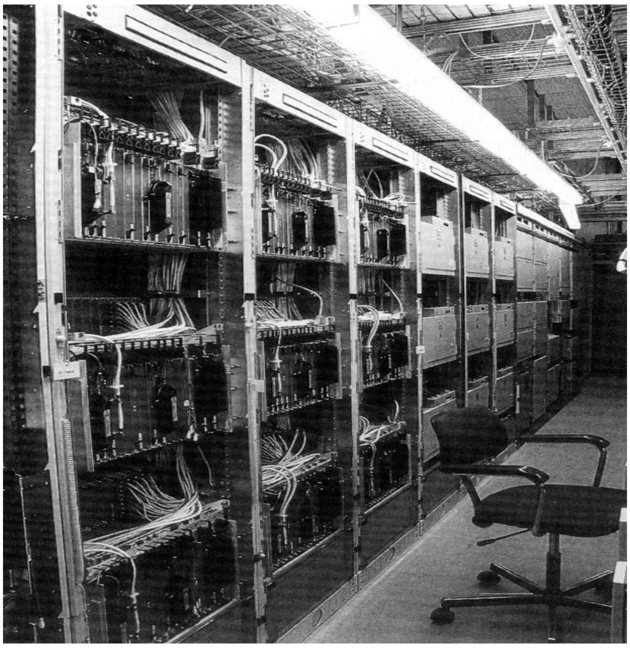

Fig 4 SDH model.

Although a primary function of the model is to provide a platform for KiloStream testing, it is also a major element in the overall IF infrastructure.

As many BT services are interconnected by KiloStream in the field, the IF models this, using the KiloStream model to provide data paths for ATM, SDH, SMDS, ISDN and other models (see Table 2).

Table 2 UK core transmission equipment models.

Equipment	Function
Various	PDH
36170 (Newbridge)	ATM VPS
X-connect (GPT)	National SDH
X-connect (GPT)	DPCN — KiloStream
X-connect (AT&T)	DPCN — KiloStream.

FeatureNet

The FeatureNet model's main component is a Nortel DMS100 Advanced Services Uni (see Fig 8).

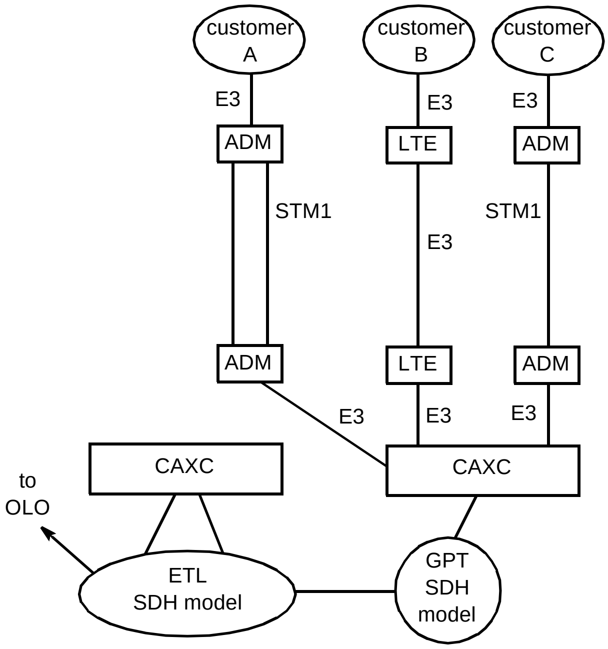

Fig 5 ATM model — international configuration.

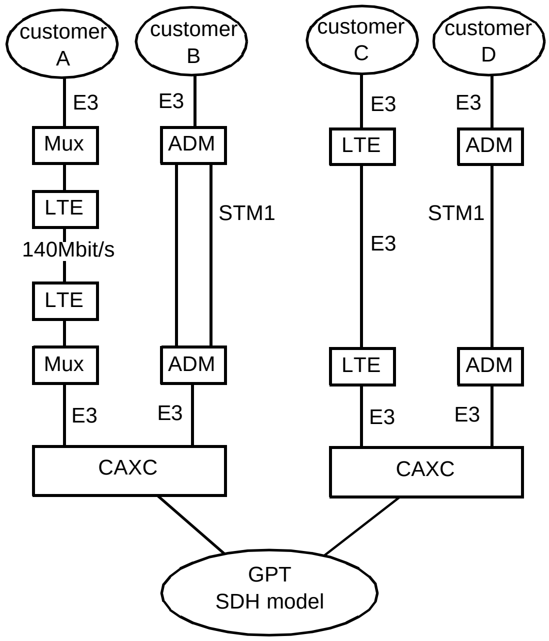

Fig 6 ATM model — inland configuration.

In common with the live network, a number of small remote units (SRUs) and remote switching centres (RSCs) provide actual service to model users. The SRUs may in turn be connected to a private branch exchange (PBX) before terminating in customers' telephones. A number of PBXs, of varying types, are available within the IF. These are shared between various projects, and are often in use by several programmes simultaneously.

As with most models, there is a high degree of interconnectivity — in this case, the central ASU is linked to core PSTN switches to test PSTN breakout, management control centres to test management access, PBXs to test customer access, and vendor-managed equipment to test interworking with multiple ASUs.

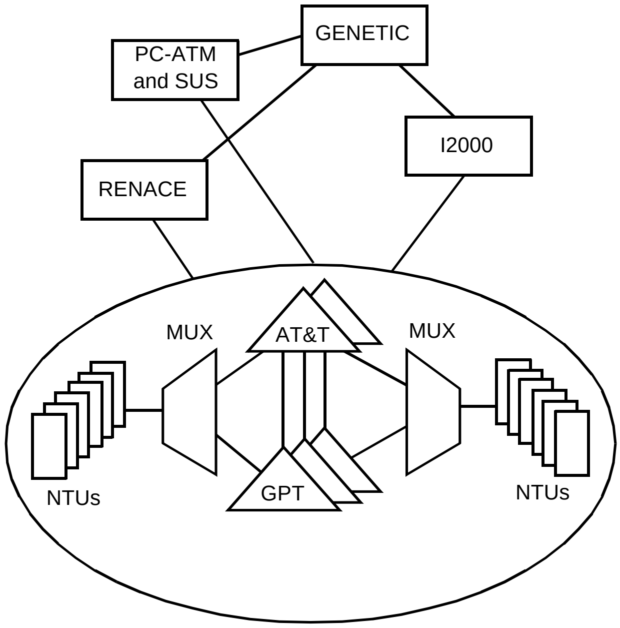

Fig 7 KiloStream model.

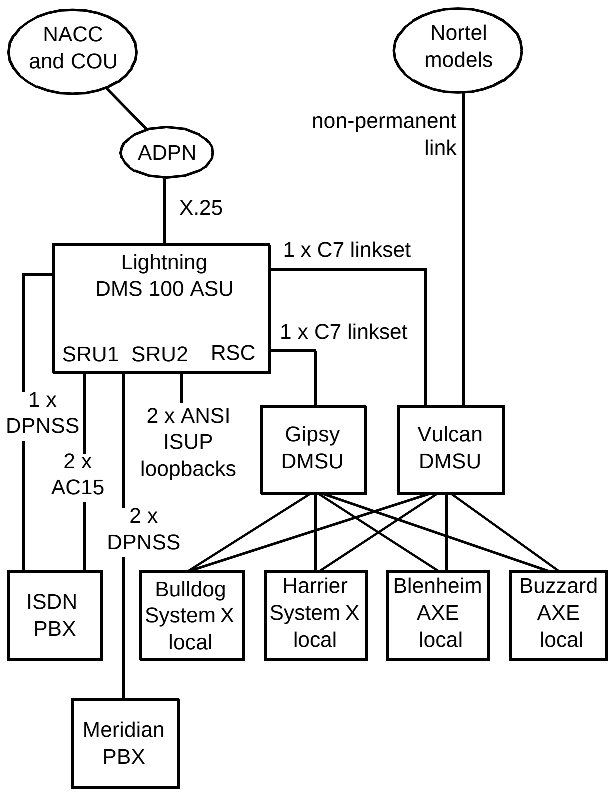

Fig 8 FeatureNet model.

Card services platform

This area is represented by the cashless services replacement system (CSRS), and its forthcoming derivative, the scalable integrated services network (ISN) [6]. CSRS is an interesting case in that the design was effectively an ‘off-the-shelf’ solution from MCI. This meant that integration effort was aimed mostly at those components specific to the UK implementation. Although this meant that development and testing time-scales could be significantly compressed, the model still needed to reflect the entire network. This model is shown in overview in Fig 9.

One of the immediately obvious aspects of Fig 9 is the complexity of such a model. The maintenance of a consistent and meaningful databuild across the entire environment occupies a significant amount of model management overhead.

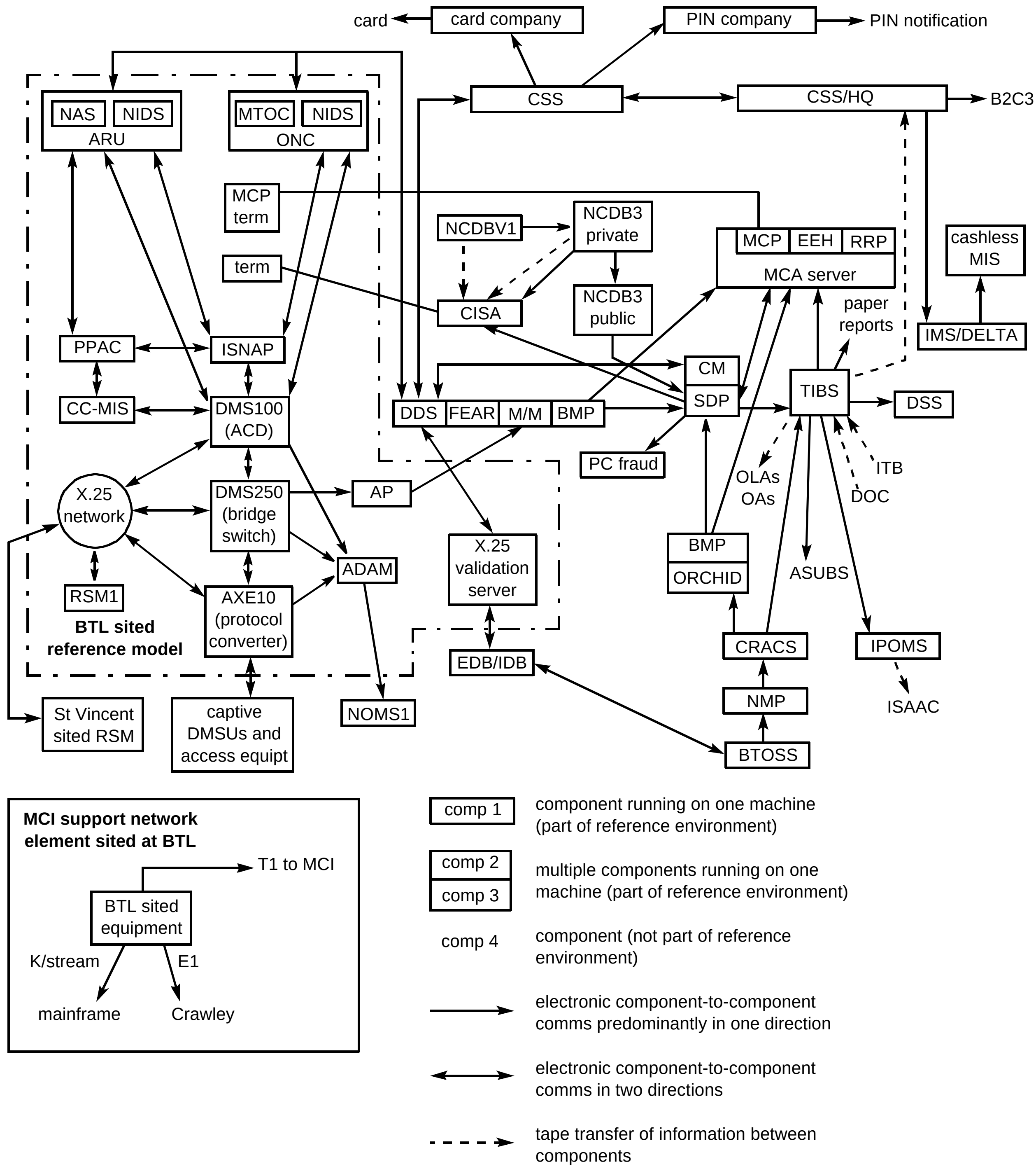

Fig 9 Card services model.

An interesting feature is that not all the equipment is located at one site — it is impractical to model some major BT systems, such as CSS, as a completely separate entity. In these cases, a separate model partition is created on an otherwise live system, and the integration facility uses these partitions as part of the overall model environment. Table 3 summarises the models.

Table 3 CSRS models.

Exchange	Name	Function
AXE10 Cashless (ETL)	Goshawk	CSRS
DMS100 (Nortel)	Electra	CSRS
DMS250 (Nortel)	Thunderbolt	CSRS

A further feature of the Cashless model has been the need to interwork closely with MCI, to the extent of having dedicated trans-Atlantic bandwidth for testing and upgrade delivery purposes. This requirement is likely to be echoed in ever-increasing volume, as joint developments become the norm.

Derived services

The Derived Services Network (DSN) was the first implementation of what is now known as an intelligent network. It is an overlay of exchanges providing a wide range of services such as 0800 Freefone, 0345 Lo-Call, 0891 Premium Rate, and Directory Assistance bureau services such as 192. The managed recorded information services (MRIS) is also part of the DSN. This offers a further range of services, such as announcements guiding callers through a menu of options, and televoting applications. DSN is modelled in the IF by an AT&T 5ESS exchange, together with the array of systems providing MRIS.

Flexible Bandwidth Service (FBS)

The Flexible Bandwidth Service effectively offers customers the benefits of a private data transmission network with highly flexible provisioning and service options.

The model environment in the IF (see Fig 10) was used not only to trial new releases of hardware and software, but was also available as a third-line support tool. This meant that BTL personnel were on a 24-hour call-out rota to rapidly diagnose problems. This in turn meant that the model was maintained in a high state of readiness at all times. Recently, the service has matured to the point where such support is now being offered by the equipment vendors. However, in the early stages of the service, the availability of the model facility was a major enabler in allowing BT to rapidly launch new features, knowing that any problems so caused could be addressed quickly within the model.

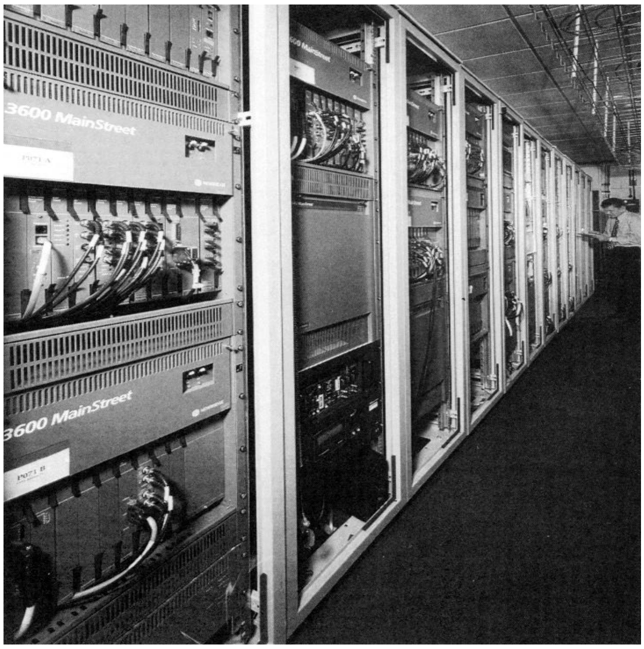

Fig 10 Flexible Bandwidth Service model.

Switched Multimegabit Data Service (SMDS)

The SMDS model is mostly used to trial new releases of hardware and software before they enter service in the field, and to test procedures, such as upgrading to new metropolitan area network (MAN) software. The model is based around two Siemens/GPT MAN clusters, managed by a Tandem fault-tolerant computer. Customer access is either via Cascade wide area network (WAN) switches, or direct from the MAN equipment.

This model originated in the BTL broadband laboratory, but once work gravitated away from development activities, it was considered sensible to logically place the model within the IF. From there, it was a small (but not inconsiderable) step to physically moving the model into the IF, which was achieved in February 1997. This migration of development-based models into integration environments is likely to re-occur, as the need to derive maximum value from equipment becomes more pressing.

UK intelligent networks

This category includes many of the relatively new services that BT has overlaid on top of its core network.

The GPT Service Control Point [7] is an evolving platform, currently used for personal number services. It is represented in the IF by one SCP (two RM600 Unix computers), and a number of associated terminals and workstations. The SCP is connected to the exchanges via time-slots in C7 links.

The interactive speech applications platform (ISAP) is the basis for a number of applications (see Table 4), all of which are tested within the IF. These applications currently include telemetry, for remote reading of meters, Call-Minder, a network-based call answering service, and call store forwarding [8], a feature of the cashless services replacement system. The ISAP is supplied by Ericsson, using VME-based hardware, and is linked to the exchanges via E1 links (Fig 11 shows the CSF configuration).

Table 4 IN application models.

Equipment	Name	Platform
GAIN 300 SCP(GPT)	Phoenix	Personal Numbering
ISAP (ETL)	Hurricane	CallMinder
ISAP (ETL)	Tempest	Telemetry
ISAP (ETL)	Whirlwind	CSF for CSRS
ISAP (ETL)	Typhoon	ACE

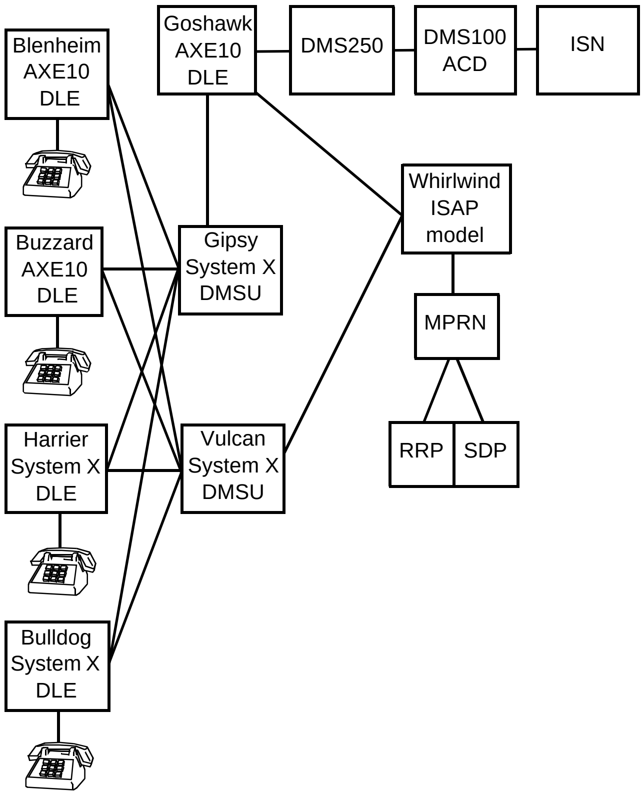

Fig 11 ISAP CSF model network diagram.

Cambridge is a major enhancement of the core PSTN network, and encompasses a wide range of new systems.

Figure 12 shows (merely) the transport network design for this model [9]. It can be seen that this model uses many of the existing facilities in the IF, but adds some additional systems. This is a common scenario for new models. In addition to the systems within the IF, the model requires links to other sites, using a variety of protocols, e.g. X.25, Frame Relay, and IP. Real links are needed for these applications, making the IF a significant customer for BT services.

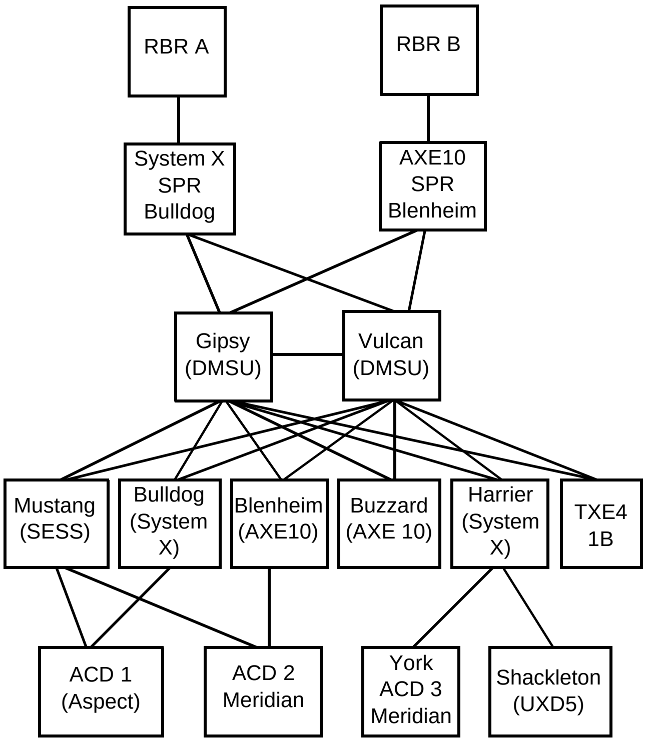

Fig 12 Cambridge model — phase 0 network design.

UK access

There are several models in place to test many different network access mechanisms [10]. The models often lie more towards the development end of the life cycle, although many are actively involved in live customer trials. For example, the telephony over passive optical network (TPON) model was used extensively during initial evaluation and trial phases, and now models the live system that has been rolled out nationally.

One-per-customer radio is modelled in the IF, and is currently undergoing trial in the network.

There are various models of systems aimed at extending the use of the existing BT access network. The digital access carrier system (DACS) provides two analogue lines over the same copper pair. Copper wideband serving section (CWSS) provides a 2 Mbit/s path over two or three copper pairs.

Several different multiplexing systems are modelled. These are at various stages of roll-out, but all offer some form of concentration of PSTN lines on to higher-order line systems.

Another aspect of access model testing is that of the front office line test systems. These are the systems in place to allow BT customer-facing people to check lines in response to reported faults. The test systems themselves are present within the IF, but external systems, such as CSS, are linked remotely. This allows full end-to-end integration testing to be performed.

Global voice

This is effectively one model encompassing the various aspects of the Concert Virtual Network Service (CVNS). The service is based around AXE-10 exchanges, of which one is present in the model. Local points of presence are represented by two business multiplexers (BMXs) and a Meridian switch, which contain a variety of different signalling systems, allowing any location in the world to be modelled in the IF. There are a significant number of management systems associated with CVNS, and the majority of these are either modelled within the IF, or logically linked to the IF from elsewhere. In particular, this includes systems in the United States, operated by MCI.

Testing is carried out on the model in a co-ordinated fashion from both sides of the Atlantic. Although the working days do not fully overlap, this is balanced by the opportunity to have use of the model for 16 hours each day, (08:00 — 13:00 UK, 13:00 — 18:00 UK and US, 18:00 — 24:00 US).

Global data

The models in the global data area are based on Concert data services. The original service was the Concert Managed Bandwidth Service (CMBS), as shown in Fig 13. This service is based on NET IDNX multiplexers, and when it was first introduced in 1994, the reference model was almost identically sized to the live network. Since then, many enhancements have been made, and the model (Fig 14) is still used to integrate new network components and services.

There is also a model of the Concert Frame Relay Service (CFRS). This area has seen the transition from a BT-only frame relay service (Global Frame Relay Service, GFRS), to the global CFRS. As the laws of probability might predict, BT and MCI had fundamentally different networks — BT using Stratacom (Cisco) and MCI using Wellfleet (Bay Networks). Much of the work on the model was related to interworking aspects between the two networks.

This model, like several other Concert models, required close liaison between development and test teams in BT and

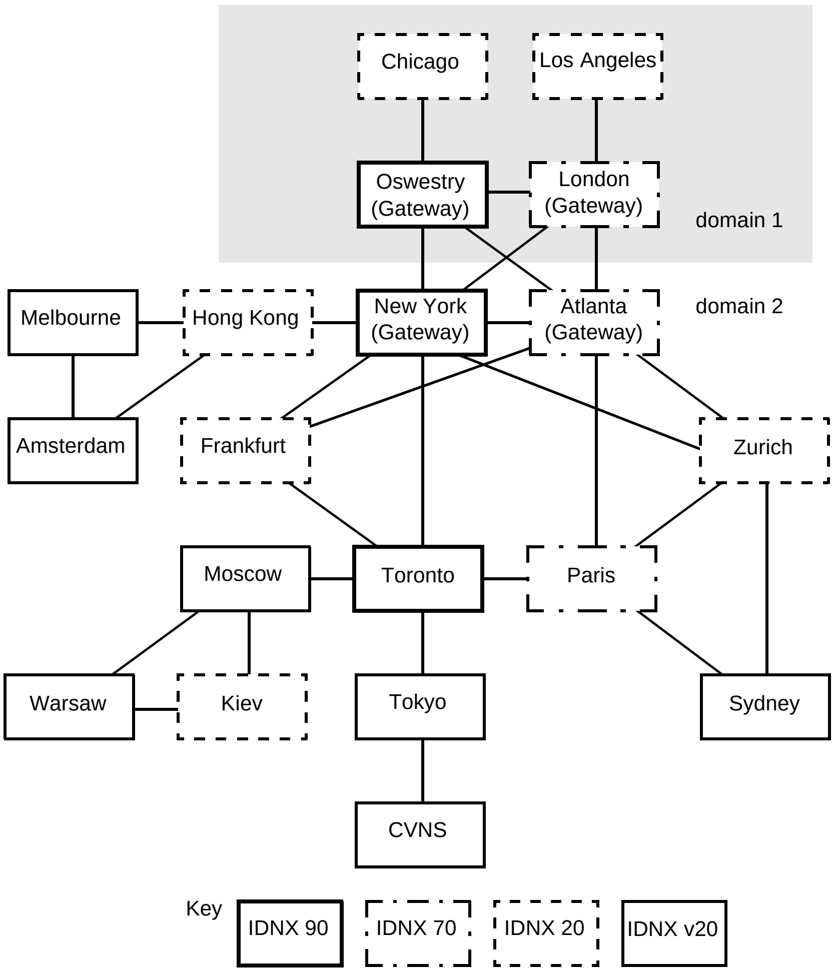

Fig 13 CMBS model topology.

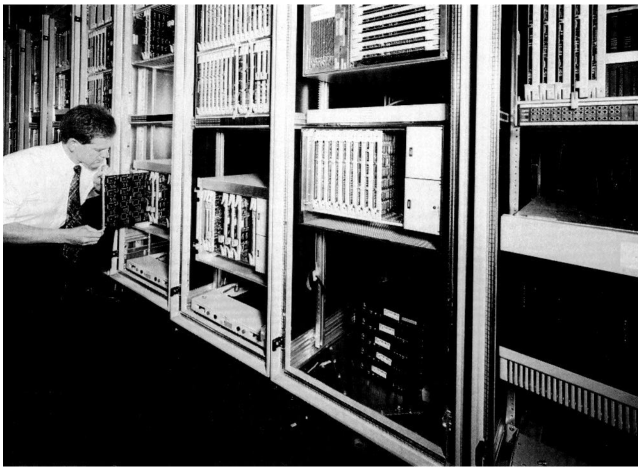

Fig 14 CMBS model.

MCI, thus fostering a good working relationship between the two companies.

Global transmission

The two main areas covered by models are the Global ATM platforms provided for core bandwidth between BT and foreign PTTs, and the Concert Global Managed Platform (GMP), used as the core platform to carry Concert's global services.

The international BT ATM model uses the same equipment as the UK ATM model, discussed earlier in this section.

The Concert GMP model is used to evaluate the best way of integrating existing services on to a common platform, as well as for normal verification, validation and testing (VV&T) activities. Much of the equipment is new to BT, so the model has provided a useful facility in which to learn its characteristics. There are two main parts to the model. One part is the ATM common platform, currently being evaluated to determine exactly how ATM will be integrated into the Concert services. Testing on the model has raised significant issues regarding the proposed implementation.

The other part of the model is the Tadiran T::DAX crossconnect switch, which currently offers an underlying circuit switched platform to the CMBS and CVNS services, using E1/E3/T1/T3 links. Table 5 summarises these models.

Table 5 Global transmission models.

Equipment	Function
X-connect (ETL)	International ATM
X-connect (Marconi)	Global ATM
X-connect (ETL)	International SDH
T::DAX (Tadiran)	Concert GMP2
ATM Switches	Concert GMP4

Alliance Partners

BT has been proactive in setting up joint ventures outside the UK. As the need to provide some means of rapidly integrating alternative networks has become pressing, the Alliance Partners Integration (API) model has been set up.

This model provides a number of predetermined configurations, which can be combined in various ways to model the majority of the network configurations that will be encountered in BT's global alliances (Fig 15).

This model is currently in the final stages of introduction, and is likely to see significant change over the coming year. However, the major components will remain relatively stable — a common core transport layer, coupled

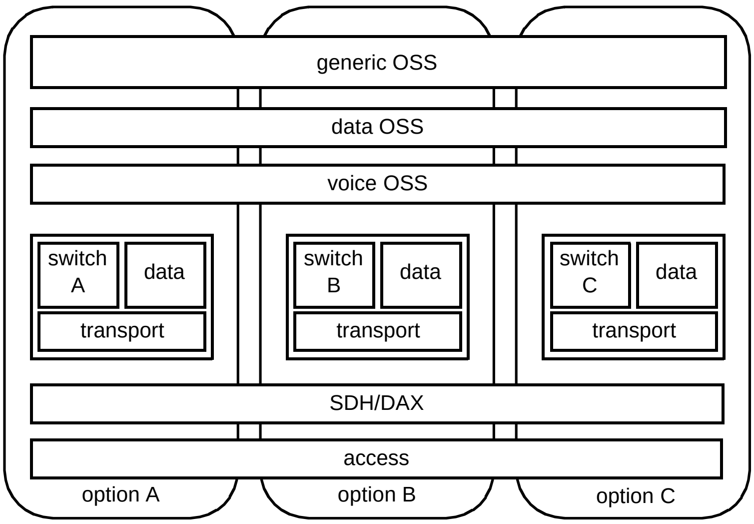

Fig 15 Alliance partners integration model.

with a range of switch types, all managed by a common OSS.

The functions of the model are typical in that they include integration testing of the data, voice, transport and OSS components of the networks. However, the model is also a vital demonstration facility — there will be no other site capable of representing all the disparate components within one integrated facility.

Multimedia and IP

The growth in interest in IP business within the last year has been phenomenal. This area is currently poorly represented in the VV&T arena, with provision of reference models being no exception. This situation is being rapidly resolved, and by mid-1997, a representative model of BT’s IP infrastructure will be in place. This model will encompass the main areas of IP connectivity, i.e. core WAN switches, WAN connectivity, routers, firewalls, and server farms.

Multimedia is a much-maligned term, meaning entirely different things to different people. Here, multimedia refers to the delivery of some form of content using computers. This is a deliberately wide definition, since new applications are being made available (almost literally) every day.

The period from concept to launch of a new service is measured in weeks, not months or years. If the new service proves unsuccessful, its lifetime may be similarly measured in weeks.

In this context, the traditional paradigm of a reference model starts to break down — no longer is it possible to set up a stand-alone model and freeze its configuration in a particular state for a few months while exhaustive testing takes place. Although the basic functions of reference models remain constant, model management systems must be able to cope with the enormous churn involved in a highly fluid design. Is this the future of reference modelling?

Additionally, this rapid churn means that new equipment (or equipment in new configurations) is not available until relatively close to service launch. This means it is impractical to build up both a development model and an integration model. Thus, both activities must take place on the same set of equipment, leading to more configuration management headaches.

3. Benefits of models

Reference models are capable of performing many different functions, depending on the requirements of any particular project at that particular time.

Multipurpose models are highly desirable, as they allow more use to be made of the equipment, thus reducing costs for each individual activity.

However, it is important to note that, although all of these functions may be possible, it does not mean that they are compatible. For example, if a model is being used by integration testers to find out what happens when the main system disks are abruptly taken off line, it is probably not a good idea to be trying to run a live trial at the same time.

In cases like this, there are two options — either dimension the model such that critical components exist in sufficient quantities to allow multiple activities to take place concurrently (in safety), or schedule the activities to operate sequentially. Every model is under pressure to deliver the maximum number of functions possible with as little equipment as possible, so calculated risks have to be taken.

One of the main functions of the model management team is to manage the interleaving of these varied (and incompatible) activities.

The activities can be classified into two groups — primary and secondary. Primary activities are sufficiently vital to warrant the existence of a model on their own merits. Secondary activities, while still vital in the overall scheme of things, will often occur on existing models, but will not, by themselves, be enough to justify a model if none already exists.

3.1 Primary reference model activities

Systems integration

This is the fundamental reason for the existence of reference models. Prior to this point in their life, most products will have existed only as a set of isolated components. While these components will have almost certainly undergone unit test, their interfaces will have been tested (at best) by stubs — simulations of connected devices. Stubs can never replace actual equipment — systems are now so complex that a reasonably accurate stub would take almost as long to build as the actual equipment itself. For example, although a new build of telephone exchange will have been thoroughly tested by its manufacturer (say Ericsson), they cannot be certain exactly how it will perform when connected to all the surrounding BT systems, never mind the myriad of other manufacturers’ equipment in the network. This end-to-end testing is vital if operational problems are to be avoided.

Hence the use of a reference model facility, where all the disparate components within a product or service can be brought together, and, for the first time, interwork as a single logical entity.

One of the best ways to test an interface specification is to ask one team to design the equipment to go on one end of an interface, and a different team to design the equipment for the other end of the interface. It can be virtually guaranteed that where any ambiguity in the interface specification exists, one team will assume a different interpretation to the other — and they will both be right. With luck, this ‘misalignment’ will be detected early in the project, but it is entirely possible that the problem will only surface when the equipment is complete, and undergoing systems integration. Where teams are within the same company, this is an embarrassing irritation, but when the teams are from different companies, a pitched legal battle can ensue.

The role of the reference model facility at this point is one of professional detachment. Since it is not part of any development team, it matters not whose fault things are, merely that there is a problem, and it can be explicitly identified and localised. Furthermore, once a way forward is identified, the reference model facility can be used to rapidly test the solution.

Of course, if all the components within a system were to originate from one supplier, then it would be entirely feasible to require that supplier to take responsibility for systems integration. The problem with this, however, is that BT would be constrained to use that supplier for all components of that system. The opportunity to select best-of-breed for each separate component diminishes, and the risk of being held to ransom by a supplier increases. In addition, given the level of interconnection within the BT network, there is always going to be a need to integrate multivendor systems at one level or another.

This system integration role is vital in protecting the network — the interdependence and complexity of modern systems means that there is an increasing risk of serious problems being introduced in new services. Without the models in the IF, it would be impossible to safely introduce these new services. The IF might legitimately be considered the ‘guardian of the network’

System enhancement

In many ways, subsequent upgrade of an existing system is much more difficult than an initial installation. Existing customers should not notice any interruption to service, existing data must be maintained, and no existing features should be lost. Note that many of the problems associated with an upgrade are related to the few minutes surrounding the actual upgrade itself — once a steady state can be achieved, half the battle is over.

Reference models are the only place where such system enhancements can be trialled without potential risk to customers. If necessary, the upgrade can be performed over and over, until all problems have been overcome. This applies not only to the deliverable itself, but also to the processes needed to support it, e.g. archiving and restoration of data, physical handling of media, and availability of accurate documentation.

Indeed, for installers working on new or unfamiliar systems, it can be immensely comforting to be able to practise on a ‘safe’ system before being unleashed on operational networks.

Support tool

Fortunately for support engineers, problems regularly occur in operational networks. Where these problems affect customers, it is essential that they are fixed, and fixed quickly. The first step is to find out what is causing the problem. If the problem is intermittent, then it can be difficult to isolate in an operational environment. A reference model can be set up to look for a specific event and, once it occurs, to capture the conditions leading up to the event and the consequences thereafter.

If the problem can be traced to a design fault, then a fix must be applied. With managers desperate for a solution, such a fix may not pass through the same rigorous checking process as a scheduled delivery (i.e. it is ‘delivered’ roughly 10 seconds after the developer has finished coding). It is far from unknown for such ‘fixes’ to introduce more problems than they solve.

With a reference model in place, all fixes can be exercised quickly but rigorously — not only to verify that they do indeed fix the problem they were designed to, but that they also do not impact upon the system in any unexpected fashion.

Of course, not all fixes require new deliverables. In many cases, support personnel can devise a workaround, either temporary or permanent. To do this, they need access to a model where they can safely try out a number of options, to see what effect they have. Indeed, this sort of activity may be going on permanently, in the never-ending struggle to improve performance and functionality.

3.2 Secondary reference model activities

Proof-of-concept model

Development teams often maintain small test beds to verify the performance of essential pieces of equipment.

Generally, these test beds are only representative in certain areas — perfectly acceptable most of the time, but sometimes the lack of a proper infrastructure renders the test bed fatally flawed. If a reference model of an existing service exists, then it is perfectly feasible to use this to supplement the development model. In this way developers are able to confirm that their designs are basically sound, thus potentially saving them from time-consuming paper evaluations or, even worse, inherent design flaws.

Product verification

It is unfair to suggest that product vendors sell ‘vapourware’. However, there have been occasions where internal development cycles have perhaps not exactly aligned with salespeople’s expectations, and hence their promises. In such circumstances, the existence of a reference model, where pre-deliveries can be verified against expected progress is a powerful incentive towards full disclosure.

Implementation check

When a model is being built ahead of a live service being rolled out, certain unforeseen issues are often raised. These issues may relate to some relatively minor inconsistency or omission in the design documentation, for example there may be a mismatch between a cable type and its intended socket. It is only at the stage where someone actually tries to implement the design for the first time that these issues are noticed. When this occurs in a model, some inconvenience is experienced, but a fix can usually be applied quickly. When it occurs in the field, perhaps in a few dozen sites, then it is a major problem. As with all developments, the earlier problems can be detected, the less cost is incurred to correct them. The role of reference models in this initial implementation stage is often overlooked, as it does not strictly form part of the integration testing. However, it is a vital pre-integration stage, and a valuable tool towards real system deployment.

Process development

Many of the operational problems associated with new services relate not to the equipment, but to the processes surrounding the equipment. Part of this involves the chicken-and-egg situation that new processes cannot be developed until the system is available, but the system cannot be available until the processes that surround it are available. In these circumstances (particularly for brand new systems), the role of the model is to provide a facility where new processes can be fine-tuned in a safe environment. One obvious example is to verify the processes related to recovery from complete system failure — not something to be practised in the field with real customers attached.

Training facility

Here is a conundrum: ‘Who trains the trainers?’ Normally, trainers will base their training packages on those developed for existing systems. These will be constantly upgraded as new features and facilities are brought on-line. However, where brand new systems are being introduced, no existing material exists. In these cases, the availability of a reference model to develop the training packages (and perhaps run initial training courses), is highly valuable.

Service trial facility

BT has traditionally been in the business of introducing major new services, on a national basis. In these cases, the service must obviously be made available by upgrading equipment already in the network. However, in the multimedia market for example, a completely new service might require very little additional equipment or network infrastructure. Multimedia markets are evolving rapidly, and no one is entirely sure how much real customer demand exists for what are often niche products. There is a need to test such services cost effectively, for some of them will never get beyond the pilot stage. Even for those services that successfully receive a full launch, there is a high likelihood of major design changes following the results of the trial service.

With careful initial design, a reference model environment is ideal for undertaking service trials. Models are under strict configuration management, allowing changes to be applied during the trial in a secure manner. Basic infrastructure is often in place, minimising set-up costs. On the downside, once a trial is running, those parts of the model concerned must be kept strictly off-limits to all other testing activities. The need to run a trial involving live customers (internal or external) demands high reliability and availability, which is incompatible with some normal testing actions, e.g. operation under extreme failure conditions.

While this activity has seen limited use so far, it is likely to become much more common in the future.

Fall-back facility

This is a function of models that remains largely unused, due to the difficulties and limitations involved. There is an obvious attraction to use a reference model as the live system fall-back — it seems a shame to build a custom fall-back solution, given that the need to use will hopefully never arise. It is much better to expand the reference model and allow it to take over in those rare cases.

Unfortunately, the needs of a fall-back facility are not always compatible with those of a reference model. To be useful, a fall-back facility must have the same (or similar) capacity. For network-based models, this almost instantly rules them out — their capacity is often severely constrained, and what capacity they have will not be in the right geographical location.

Also, models will be in a particular testing configuration, and will take time to reconfigure to a 'fall-back build'. Depending on the model involved, such reconfiguration may take so long as to make the idea unworkable.

Finally, reference models are inherently not as secure as their live network counterparts. If live traffic is to be carried over a model, then access and general security arrangements must be specially provided, to a much higher degree than normal for a model environment. This is not to say that a model is insecure, just that its normal mode of operation is at odds with the secure conditions found in the network.

Although it was stated above that models cannot be used normally for fall-back, in some cases this is far from true.

Mobile exchanges are held in readiness by BT to rapidly provide service in cases where exchanges are lost. These mobile units are, by definition, unused most of the time. Several of these units are in use at BTL as reference models, with the expectation that they may be re-deployed at a few hours' notice.

One explicit example of a model being used for fall-back was when the CVNS network was being upgraded. In the live network, either one of two computer systems could support the entire network. However, when one was taken out of service to be upgraded, there was a risk that the remaining system could suffer a failure. Thus, for the duration of the upgrade, the reference model was configured as a fall-back site. In this instance, the duration of the fall-back was strictly controlled, and it was entirely feasible for the model to provide a fall-back facility.

Bid support

There have been many occasions where models have been used to verify that a particular configuration for a particular customer is valid. This may be needed for internal confirmation, or to demonstrate to a customer that BT's claims are valid.

These configurations are often (by definition) untried, and cannot be introduced into any associated live networks due to the risks involved. In such cases, a reference model environment is perfect, and is frequently instrumental in demonstrating BT's commitment to the bid. It is obvious to the customer that BT actually tests its products before use, and is able to provide a comprehensive after-sales back-up.

One example was the successful work that was done to support the bid for the new Halifax network. This involved components from several models being brought together at very short notice to determine exactly what network functionality could be offered. The results were surprising to the customer, and the way in which BT was able to provide demonstrations of actual network configurations was obviously a major differentiator between the BT bid and the competition.

Demonstration facility

Merely the existence of the IF, with its range of models, sends the clearest possible signals to potential and existing customers that BT is serious about integration testing. Thus, in terms of a demonstration tool, its greatest asset lies in just being there. Some models are conceived from the outset as demonstration aids — the interactive TV reference model was not just an integration facility, it looked the part. Initially, most models were implemented as efficiently as possible, with ergonomics taking a distant second place. However, it is now recognised that being able to drive home BT's commitment to testing is a valuable selling point, and models deserve to be designed accordingly. Of course, most visitors are accepting that they are being shown a working environment, and many are obviously delighted to see real equipment, and not just another paper presentation. These reference models are constantly in use for both formal visits and informal presentations.

4. Model management

4.1 Overview of model management

The models are managed by the network models and support unit, which is within the Systems Integration department. The role of the unit falls into three main categories — introduction of new models, maintenance of existing models, and operation of existing models.

There are several benefits behind having a dedicated team of people to manage models. Management issues tend to be common across models, so there are cost advantages in having a team with expertise in those issues. A large pool of models can be managed in a much more consistent way than a series of individual models, bringing improved response times and a wider coverage of integration scenarios.

Thanks to the common processes in use across the models, series of metrics have been gathered over the last few years. These metrics have been analysed to provide synthetics, or derived results, allowing increasingly accurate estimation of future workloads, based on anticipated demand [11].

4.2 Model introduction

The primary function of the models in the IF is integration testing. For this testing to be completely valid, the behaviour of the model must be identical to the corresponding live network. As this is rarely possible in the real world, the model is always a compromise between what is desired and what is practical. Obviously this means there is a risk that some tests which are successful in the model environment will fail when that situation occurs in the real network. Part of the skill in integration testing is to ensure that these risks are minimised, and part of that skill is to design a suitable model.

The starting point for a model design is always the live network. Ideally, the model would be identical, and where components are few and inexpensive, then this may be possible. In most cases, components are neither, and a balanced judgement must be made how best to minimise any discrepancy between the model and the operational equivalent.

In cases where many instances of a single component exist, a model will typically be equipped with as few of that component as necessary. This means absolute performance testing is not possible, although load conditions may usually be extrapolated based on some combination of metrics.

Full redundancy may not always be provided in a model other than testing fall-back procedures, as there is insufficient reason to justify the expense.

Where lower performance components do not significantly affect the functionality of a model (e.g. lower speed CPUs), a model can be scaled down cost effectively. Again, some extrapolation will be required before any performance measurements from the model can be applied to the live network.

4.3 Model maintenance

Model maintenance is defined as those things that would have to take place, even if no users were present. Thus equipment will still break down, air-conditioning will still be required, and routine back-ups must still be taken. These tasks are an essential part of keeping the reference models ready for testing at all times.

4.4 Model operation

A major headache in the operational phase of model management is trying to accommodate a number of different (simultaneous) users on the same equipment. This needs careful scheduling, and a thorough understanding of not only the equipment within the model, but also the scope of testing demanded. There is always a risk that concurrent users may interfere with each others' testing, but the alternative, of only allowing one set of testing to be carried out at any given time, would lead to impossibly long time-scales. The final decision, on what parallel testing is possible, always lies with the model managers. They also have the responsibility of deciding how the model can be best partitioned to meet the varying demands, bearing in mind the opposing demands of costs and time-scales. The expertise to make these decisions is not easily gained, especially with the wide range of models present in the IF.

The other most significant difficulty in model operation is that of configuration management. Each model may have dozens of separate configurable items, each of which must be maintained in a known state, otherwise test results gathered may be worthless.

Effective scheduling and configuration management could not be accomplished without the aid of a comprehensive booking system, allowing a high degree of automation in the scheduling and job allocation processes.

The ability to generate traffic for the models cannot be overlooked. It is futile to provide a model, and not be able to load it with realistic traffic. Many testers, e.g. the Martinet PSTN call generator or the Smartbits IP traffic generator, are available in the IF to effectively exercise the models.

5. MCI reference modelling approach

BT's integration facility and MCI's I-Lab are two of the largest reference model environments in the world. In essence, their approaches to reference modelling are very similar. In one way, this is surprising, given the differing backgrounds of the companies — MCI is a recent company, with none of the historical legacy that BT inherited from public service days. In another respect, the similarity is perhaps understandable — both companies are committed to high-quality services, and are the two largest telcos to have committed to a multi-supplier policy, partly to permit best-of-breed solutions.

Differences come down more to implementation than strategy. The only significant differences are in scale and coverage. MCI has a relatively homogenous network, significantly biased towards trunk traffic. BT has a wide-ranging network, covering trunk, access, and a cornucopia of overlying services. Additionally, MCI tend to implement upgrades in very small doses, but very regularly. BT tends to have fewer, but more major releases. It is this increased churn that requires MCI to have a larger model, in relative terms — narrow range but deep coverage.

6. The future of reference modelling

What does the future hold for reference modelling? Few things are certain, but that reference models will continue to play a central and growing role in BT's integration strategy for some time to come, is one of those certainties. There is simply no alternative. However, a number of changes can be expected, some of which are already starting to make their presence felt.

Traditionally, reference models were managed as one homogenous facility. For example, this meant that when a new model was introduced, the rest of the facility would be modified as necessary to ensure that maximum benefit was accrued from the new model. Today, the move towards trading units means that each new model is targeted at a particular service, and funding is provided to test that service. There is a risk that, without careful management, the IF will become less of an integrated facility, and more of a collection of models used on particular projects.

The inexorable compression of development (and hence testing) time-scales is certain to modify the role of models. Configuration management will become even more important as the frequency and complexity of new software and hardware builds are reflected in the release processes. If reference models are to remain an essential ingredient in the delivery process, they must continue to evolve in the same way as the network, i.e. the emphasis on rapid delivery must not affect the quality of product delivered.

Automated testing tools will become a much more integral part of reference models. This is perhaps the only way that full-scale testing will remain possible, given the limitations of resources, in terms of both people and time available. Automated testing is already being used in the IF, with extensive use on many of the core network switch models, and further expansion to follow [12]. The concept is very well established in the MCI I-Lab, especially for regression testing.

There are going to be many opportunities for BT and MCI to share best practice. Joint projects will become even more prevalent, and the creation of a single company will accelerate this process, to the benefit of both parties.

7. Conclusions

Reference models are an integral part of the VV&T process. While it is unfair to say that a good model will result in good testing, it is certainly true that a poor model will seriously limit the useful results that can be produced.

As models become more cost-constrained, it is essential that maximum value be extracted from them. This has already led to multi-function models, with VV&T being merely the main activity — perhaps in the future, not even that will be true.

The usefulness of reference models cannot be understated — without them, BT would either be releasing unproven products into the market-place, or would be reliant on suppliers' facilities. The existence of the IF provides BT with a vital differentiator, allowing absolute confidence in the quality of released products.

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